INCH-POUND
MIL-PRF-1/877E
w/AMENDMENT 1
4 October 2021
SUPERSEDING
MIL-PRF-1/877E
27 March 2012

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER, TYPE 100TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Triode.

See figure 1.

Weight: 4-ounces (113.4 grams) nominal.

Mounting position: Vertical.

ABSOLUTE RATINGS: C Telegraphy

Parameter: Unit:	F1 MHz	Ef V	Eb V dc	Ec V dc	Ib mA dc	Ic mA dc	Pp W
Maximum: Minimum:	40 ---	5.25 4.75	3,000 ---	-210 ---	225 ---	50 ---	100 ---
Test conditions:	---	5.0 V ac	2,500	Adjust	40	---	---

See footnotes at end of table I.

GENERAL:

Qualification: Not required.

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

AMSC N/A
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FSC 5960

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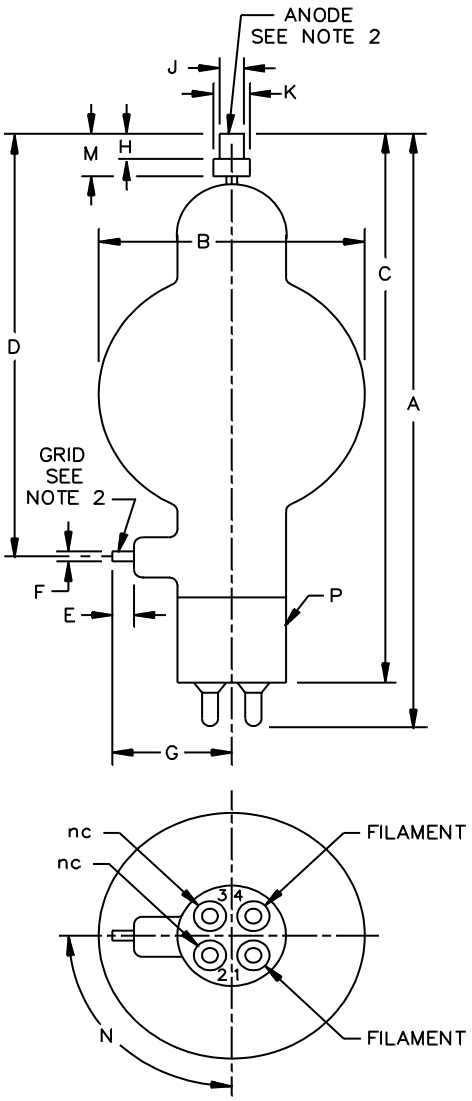
TABLE I. Testing and inspection.

Inspection	Method MIL-STD-1311	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1							
Filament current	1301	---		If Ec(1)	5.8	6.6	A ac
Electrode voltage (1) (grid)	1261	---		Ec(2)	-34.0	-46.0	V dc
Electrode voltage (2) (grid)	1261	---	Eb = 14.0 kV dc; Ec/Ib = 1.0 mA dc	Ic	---	-550	V dc
Total grid current	1266	1/		Isg	---	-10	µA dc
Primary grid emission	1266	---	Ic = 100 mA dc; t = 15 seconds; Ef = 5.5 V ac	is	---	-500	µA dc
Peak emission	1231	---	eb = ec = 2,500 v		3.0	---	a
Conformance inspection, part 2							
Amplification factor	1316	---		Mu	34.0	42.0	---
Power oscillation	1236	---	F = 8 MHz; Eb = 3,000 V dc; Ib = 150 mA dc	Po	270	---	W (useful)
Direct-interelectrode capacitance	1331	---		Cgp	1.7	2.3	pF
				Cin	2.5	3.4	pF
				Cout	---	0.45	pF
Conformance inspection, part 3							
Low-frequency vibration	1031	2/	No voltages applied	---	---	---	---
Shock	1042	2/	MIL-STD-202-213	---	---	---	---
Life test	---	---	Group C; power oscillation; t = 500 hours	---	---	---	---
Life-test end points:	---	---					
Peak emission	1231	---		is	2.5	---	a
	1266	---		Isg	---	-500	µA dc

1/ This test is to be the first test performed at the conclusion of the holding period.

2/ This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of zero.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	7.250	7.750	184.15	196.85
B	---	3.188	---	80.98
D	5.000	5.500	127.00	139.70
E	.250	---	6.35	---
G	1.250	1.500	31.75	38.10
N	80°		100°	
Conformance inspection, part 3 (see note 1)				
C	6.625	7.125	168.28	180.98
F	.055	.061	1.40	1.55
J	.350	.365	8.89	9.27
P	Base: A4-10 (EIA)			
Reference dimensions				
H	.328		8.33	
KD	.500		12.70	
M	.563		14.30	

NOTES:

1. These dimensions shall be checked annually with the following sampling plan:

n1 = 4	c1 = 0	where c2 represents the total failures for the
n2 = 4	c2 = 0;	first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become conformance inspection, part 2, for three successful consecutive submissions, at which time the test may revert to the conformance inspection, part 3 basis.

2. Heat radiating connectors HR-2 (grid) and HR-6 (anode) or equivalent, are recommended for all applications.

FIGURE 1. Outline drawing of electron tube type 100TH.

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Referenced documents. In addition to MIL-PRF-1, this specification sheet references MIL-STD-1311 and MIL-STD-202-213.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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